

Nevette Bailey, Ph.D.

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Senior software engineer with 6 years of production engineering experience in monetization and payments systems at Etsy and as founding engineering leader at a high-growth retail startup. Cross-domain background in academic research and professional ballet brings the same precision and systems thinking to engineering.

EXPERIENCE

Independent Research Engineer – Ethical ML Systems

First On Screen: Representation Tracker | New York, NY

JULY 2025 – PRESENT

- Designed and built an end-to-end ethical ML system for tracking media representation using self-identification-only data, explicitly avoiding appearance-based classification.
- Implemented a multi-source AI-assisted data pipeline integrating TMDB, Wikipedia, web search, and the Claude API to extract explicit self-identification statements with source attribution and confidence scoring.
- Architected a three-layer FastAPI backend (routers, business logic, services) with human-in-the-loop verification workflows and full data provenance.
- Built Python services and orchestration logic for create-or-merge workflows, preventing duplicate submissions while maintaining full audit trails across pipeline runs.
- Designed a PostgreSQL / Supabase schema using UUIDs, JSONB, and array fields to support intersectional identities, verification states, and research-grade traceability.
- Implemented background job processing (Celery + Redis) for batch ingestion and long-running workflows with progress tracking and retry safety.
- Built admin review tooling and APIs enforcing AI safety principles: AI suggests, humans verify; no automated writes to production data.
- Prototyped statistical evaluation methods (bootstrap confidence intervals, bias detection, effect sizes) to quantify uncertainty and fairness in ML-assisted analysis.

Vice President of Engineering + Technology (Founding Engineer)

Gotham | New York, NY

MARCH 2024 – JULY 2025

- Designed and scaled e-commerce infrastructure to support multi-location expansion, maintaining 99.9% uptime during peak traffic periods.
- Identified a critical single point of failure tied to a single third-party provider and built a backup POS system supporting multiple payment types through independent processors, with day-of inventory snapshot tracking, preserving full sales capability during high-volume outages.
- Built and owned SEO technical implementation that drove 536% year-over-year organic traffic growth and measurably improved top-of-funnel revenue.
- Architected and delivered POS/online inventory sync system enabling real-time stock accuracy across physical and digital channels.
- Established CI/CD pipelines and deployment processes across a team of 5 engineers that reduced delivery timelines by 50% and defect rate by 40%.
- Built a centralized data warehouse supporting executive analytics and accelerating production iteration cycles.

Senior Software Engineer

Etsy | Brooklyn, NY

MARCH 2020 – JANUARY 2024

- Designed and launched a new Etsy Ads credit system that eliminated seasonal support tickets and improved seller trust.
- Built campaign management features that increased ad adoption by 32% and drove platform revenue growth.
- Partnered with data science to evolve seller-side A/B experimentation, improving targeting precision and measurement accuracy.
- Launched a risk mitigation project within the payments platform that captured \$700k in revenue.
- Mentored 11+ engineers, led multiple launches with zero post-release incidents, and modeled inclusive engineering practices.
- Co-chaired Etsy's Black ERG (Bridge), growing internal mentorship programs and leading equity-centered technical initiatives.
- Acted as a *de facto* technical lead on seller-experience improvements, shaping architecture, prioritization, and code quality across teams.
- Promoted twice in three years from Software Engineer to Senior Software Engineer; recognized as top peer-nominated recipient of the 2021 Etsy "We Lead With Optimism" company-wide award.

Doctoral Researcher

Columbia University | New York, NY

JULY 2011 – MAY 2019

- Led multi-year research into translational fidelity mechanisms, designing experiments and custom analysis tools to study protein synthesis errors with medical and scientific applications.
- Selected for NDSEG Predoctoral Fellowship (2013, Department of Defense-funded) and Ford Foundation Dissertation Fellowship (2016), and supported by NIH and NSF grant programs; presented findings at international conferences.
- Co-authored multiple peer-reviewed publications; mentored junior researchers and managed lab operations in a federally regulated environment.

SKILLS

Languages & Frameworks: Python, TypeScript, JavaScript, FastAPI, Node.js, React, Redux Toolkit, Vue/Nuxt, GraphQL, SQL, HTML/CSS, PHP, Dart, Flutter, Electron

Backend & Data Systems: PostgreSQL, Supabase, JSONB, REST APIs, Background Jobs (Celery, Redis), Data Modeling, Schema Design, Data Pipelines

Infrastructure & DevOps: GCP, AWS, Cloud-hosted Application Deployments, Environment-Aware Logging, Scalable Architecture

Testing & Tooling: Pytest, Jest, PHPUnit, Jasmine, Enzyme, Cypress, React Testing Library, Git, CI/CD

AI & Applied ML Systems: Human-in-the-Loop ML Systems, LLM-Assisted Pipelines (Claude), AI Safety-Aware Design, Confidence Scoring, Source Attribution, Agentic Development (Cursor, Claude, ChatGPT)

Leadership & Product: Growth Engineering, Monetization Systems, Experimentation Platforms, Technical Strategy, Agile Development, Mentorship

EDUCATION

Columbia University, New York, NY

Ph.D. May 2019, M.Phil October 2015, M.A. February 2013 (*Chemistry*)

Northwestern University, Evanston, IL

M.S. June 2011, B.A. June 2011 (*Chemistry, French minor*)

Grace Hopper Program at Fullstack Academy, New York, NY

Software Engineering Immersive October 2019